COMP SCI/SFWR ENG 4E03: Probability Review

A Crash Course in Probability

- should see most of this material in greater depth in STATS course, if you have not done so already
- will do enough here to bootstrap to start delving into performance models
- Do not hesitate to ask for more details of any concepts!

Samples and Sample Space

- Suppose that we have a data centre that has three servers (*A, B, C*)
- At any (random) point in time, we look to see which servers are up (working) and which servers are down (failed)
- *Experiment*: which servers are up, which are down?
- *Sample space (Ω):* set of all possible samples/outcomes:

$$\Omega: \{(w,w,w),(w,w,f),(w,f,w),(w,f,f),(f,w,w),(f,w,f),(f,f,w),(f,f,f)\}$$

where the components indicate if *A, B, C* are working (*w*) or failed (*f*), respectively

Events

- An event, *E*, is any subset of the sample space, *Ω*
- E_1 corresponds to Server *A* failed:

$$E_1 = \{(f, w, w), (f, w, f), (f, f, w), (f, f, f)\}$$

- E_2 corresponds to at least two servers failed
- Unions and intersections of events are also events:

$$E_1 \cap E_2 = \{(f, w, f), (f, f, w), (f, f, f)\}$$

Interpretation?

- Complement of an event is also an event:

$$E_2^c = \{(w, w, w), (w, w, f), (w, f, w), (f, w, w)\}$$

Interpretation?

Probabilities

- $0 \leq P\{E\} \leq 1$: probability of event *E* occurring
- Suppose that we have:

$$P\{(w,w,w)\} = 0.8500$$
$$P\{(w,w,f)\} = 0.0500$$
$$P\{(w,f,w)\} = 0.0600$$
$$P\{(w,f,f)\} = 0.0030$$
$$P\{(f,w,w)\} = 0.0300$$
$$P\{(f,w,f)\} = 0.0030$$
$$P\{(f,f,w)\} = 0.0030$$
$$P\{(f,f,f)\} = 0.0010$$

- $P\{E_1\} = 0.0300 + 0.0030 + 0.0030 + 0.0010 = 0.0370$
- $P\{E_1 \cap E_2\} = 0.0030 + 0.0030 + 0.0010 = 0.0070$

Unions

$$P\{E \cup F\} = P\{E\} + P\{F\} - P\{E \cap F\}$$

The last term accounts for the fact that samples in both *E* and *F* have been counted twice in the first two terms. For example, (*f,w,f*), (*f,f,w*), and (*f,f,f*) appear in both E_1 and E_2.

Theorem

$$P\{E \cup F\} \leq P\{E\} + P\{F\}$$

Equality when $E \cap F = \emptyset$, in other words when *E* and *F* are mutually exclusive.

Example. E_3 corresponding to all servers working and E_4 corresponding to all servers failed are mutually exclusive.

Conditional Probability

Definition. The conditional probability of event *E* given event *F* is written as $P\{E|F\}$ and is given by the following:

$$P\{E|F\} = \frac{P\{E \cap F\}}{P\{F\}}$$

Note that the denominator renormalizes over the new sample space (*F*), and the numerator restricts to samples that are in *E* and in the new sample space.

Conditional Probability - Example

We wish to compute the probability that *A* is failed given that exactly one server is failed.

$$F = \{(f, w, w), (w, f, w), (w, w, f)\}$$
$$E = \{(f, w, w), (f, f, w), (f, w, f), (f, f, f)\}$$

$$\frac{P\{E \cap F\}}{P\{F\}} = \frac{0.0300}{0.0300 + 0.0600 + 0.0500} = 0.2143$$

Independent Events

Events *E* and *F* are independent if

$$P\{E \cap F\} = P\{E\}P\{F\}$$

The probability of *E* occurring is not affected by whether *F* occurs or not.
Also captured in conditional probability:

$$P\{E|F\} = \frac{P\{E \cap F\}}{P\{F\}} = \frac{P\{E\}P\{F\}}{P\{F\}} = P\{E\}$$

Independent Events – Example 1

Are the events that server *A* is failed (*E*) and server *B* is failed (*F*) independent?

$$E = \{(f, w, w), (f, w, f), (f, f, w), (f, f, f)\}$$
$$F = \{(w, f, w), (f, f, w), (w, f, f), (f, f, f)\}$$

$$P\{E \cap F\} = 0.0030 + 0.0010 = 0.0040$$
$$P\{E\} = 0.0300 + 0.0030 + 0.0030 + 0.0010 = 0.0370$$
$$P\{F\} = 0.0600 + 0.0030 + 0.0030 + 0.0010 = 0.0670$$

Using the conditional probability test,

$$\frac{P\{E \cap F\}}{P\{F\}} = \frac{0.0040}{0.0670} = 0.0597 \neq P\{E\}$$

so events *E* and *F* are not independent

Independent Events – Example 2

Suppose instead that we have:

$$P\{(w,w,w)\} = 0.7866$$
$$P\{(w,w,f)\} = 0.0414$$
$$P\{(w,f,w)\} = 0.0684$$
$$P\{(w,f,f)\} = 0.0036$$
$$P\{(f,w,w)\} = 0.0874$$
$$P\{(f,w,f)\} = 0.0046$$
$$P\{(f,f,w)\} = 0.0076$$
$$P\{(f,f,f)\} = 0.0004$$

$$P\{E \cap F\} = 0.0076 + 0.0004 = 0.0080$$
$$P\{E\} = 0.0874 + 0.0046 + 0.0076 + 0.0004 = 0.1000$$
$$P\{F\} = 0.0684 + 0.0076 + 0.0036 + 0.0004 = 0.0800$$

Independent Events – Example 2 (cont'd)

Using the conditional probability test,

$$\frac{P\{E \cap F\}}{P\{F\}} = \frac{0.0080}{0.0800} = 0.1000 = P\{E\}$$

so events *E* and *F* are independent

Independent Events – Example 2 (cont'd)

Where did my numbers come from?

- The probability that server *A* is working is 0.90 (the probability that it is failed is 0.10)
- The probability that server *B* is working is 0.92 (the probability that it is failed is 0.08)
- The probability that server *C* is working is 0.95 (the probability that it is failed is 0.05)

If we assume each server's status is independent of the others:

$$P\{(w,w,w)\} = (0.90)(0.92)(0.95) = 0.7866$$
$$P\{(w,w,f)\} = (0.90)(0.92)(0.05) = 0.0414$$

We can continue in this manner to get the rest of the values.

Bayes Law

Suppose that we know $P\{F|E\}$ but we would like to know $P\{E|F\}$.

Theorem.

$$P\{F|E\} = \frac{P\{E|F\}P\{F\}}{P\{E\}}$$

Bayes Law - Example

Someone is tested for a (rare) disease. Suppose that the testing methods for the disease are correct 99 percent of the time (if one has the disease, the test is positive 99 percent of the time, if one does not, the test is negative 99 percent of the time). Suppose that the disease occurs in the general population at the rate of one every 10,000 people.

What are the chances that someone testing positive actually has the disease?

Bayes Law - Example

- Event *A*: one has the disease
- Event *B*: one tests positive

$$P\{A|B\} = \frac{P\{A\}P\{B|A\}}{P\{B\}}$$

We have $P\{A\} = 0.0001$, $P\{B|A\} = 0.99$, but do not have $P\{B\}$. To calculate this probability, use the *Law of Total Probability*.

Bayes Law - Example

$$P\{B\} = P\{B|A\}P\{A\} + P\{B|A^c\}P\{A^c\}$$
$$= (0.99)(0.0001) + (0.01)(0.9999)$$

So,

$$P\{A|B\} = 0.0098$$

Random Variables

- A random variable X (capital letters) is a real-valued function of the outcome of an experiment
- Examples:
  1. The sum of the rolls of two dice
  2. Number of requests for a website by some time t
  3. The time between arrivals to a website
  4. The CPU requirement (in seconds) of an HTTP request

Random Variables

- 1 and 2 are *discrete*, 3 and 4 are *continuous*
- For 2, if N is the associated random variable, then we can define events such as "the number of arrivals by time t is less than or equal to five" and "the number of arrivals by time t is between 1 and 3". So, we can use everything that we have discussed so far with respect to such events.

Distributions

- We would like a compact way of defining the probabilities of events.
- We will use the (cumulative) distribution function (distribution, c.d.f):

$$F_X(x) = P\{X \leq x\}$$

- We will use *F(x)* if the variable is clear

Continuous Random Variables

- for quantities that take on values on a continuum – time is an important one for us
- $F(x)$ is continuous
- Associated *density function:*

$$f_X(x) = \frac{dF_X(x)}{dx}$$

- $P\{a \leq X \leq b\} = F_X(b) - F_X(a) = \int_a^b f_X(x)dx$
- $\int_{-\infty}^{\infty} f_X(x)dx = 1$

Discrete Random Variables

- Probability mass function (p.m.f.) gives the probabilities that the random variable equals values in a discrete set (its range of possible values)

$$p_X(i) = P\{X = i\}$$

- $P\{j < X \leq k\} = F_X(k) - F_X(j) = \sum_{j<i\leq k} p_X(i)$

- $\sum_i p_X(i) = 1$

Summary Statistics - Mean

Expected value (or mean) is given by (discrete, continuous):

$$E[X] = \sum_{i} i p_X(i)$$

$$E[X] = \int_{-\infty}^{\infty} x f_X(x) dx$$

- Also sometimes called the average, but we will reserve the word average for another purpose (will become clear later)
- Note that the expectation does not have to be an outcome – the roll of a die is a classic example, $E[X] = 3.5$

Higher Moments

In general, one can compute the *k*th moment:

$$E[X^k] = \sum_i i^k p_X(i)$$

$$E[X^k] = \int_{-\infty}^{\infty} x^k f_X(x)\,dx$$

Important for us are $k = 1$ (mean), $k = 2$

Variance

An important summary statistic is the *variance*, which measures the expected deviation from the mean, i.e., how spread out the distribution is:

$$Var(X) = \sigma_X^2 = E[(X - E[X])^2]$$

Related to the variance is its square root, the *standard deviation*, σ_X. We will use the following fact to compute the variance:

$$E[(X - E[X])^2] = E[X^2] - 2E\big[XE[X]\big] + (E[X])^2$$
$$= E[X^2] - 2E[X]E[X] + (E[X])^2$$
$$= E[X^2] - (E[X])^2$$

So, variance can be calculated from the first and second moments.

Example

Let

$$f(x) = kx(100 - x), \qquad 0 \leq x \leq 100$$

and 0 otherwise be the density of the lifetime of a component (in days).

1. Find the probability that the lifetime is between 9 and 90 days.
2. Find the mean lifetime.

Some Useful Distributions – Geometric

$$p_X(k) = q(1-q)^{k-1}, \qquad k = 1,2,\ldots$$

where *q* is the probability of success, *1-q* is the probability of failure, and $p_X(k)$ is the probability that success first happens on the *k*th trial. Arises in applications like testing: if we do independent tests, how many tests must be performed before success.

With a bit of work, one can show that

$$E[X] = \frac{1}{q}$$

$$Var(X) = \frac{1-q}{q^2}$$

Some Useful Distributions – Poisson

The Poisson distribution has one parameter, λ, and has p.m.f.

$$p_X(k) = \frac{e^{-\lambda}\lambda^k}{k!}, \qquad k = 0,1,2,\ldots$$

Sums to one?

$$\sum_{k=0}^{\infty} e^{-\lambda}\frac{\lambda^k}{k!} = e^{-\lambda}\sum_{k=0}^{\infty}\frac{\lambda^k}{k!} = e^{-\lambda}e^{\lambda} = 1$$

Also,

$$E[X] = \lambda$$
$$Var(X) = \lambda$$

(The last two results require a bit of work.)

Some Useful Distributions – Poisson

Simeon Poisson (French statistician) first proposed the model, for things like wrongful convictions. Bortkiewicz modelled death by horse kicks in the Prussian army. We well see that it has up-to-date uses!

Some Useful Distributions – Uniform

- The uniform distribution, denoted by $U(a, b)$, has density

$$f(x) = \frac{1}{b-a}, \qquad a \leq x \leq b$$

and 0 otherwise.

- This means that the random variable is "equally likely" to take values anywhere between a and b. This **does not mean** that $P\{X = i\} = P\{X = j\}$, although this is trivially true. It does mean, for example, $P\{0.1 \leq X \leq 0.2\} = P\{0.3 \leq X \leq 0.4\}$ (as long as $a \leq 0.1$ and $b \geq 0.4$).
- An important special case is $U(0,1)$, as many built-in random number generators produce samples from this distribution.

Some Useful Distributions – Uniform

$$E[X] = \int_a^b \frac{x}{b-a} dx = \frac{1}{b-a} \frac{x^2}{2}\bigg|_a^b = \frac{b^2 - a^2}{2(b-a)} = \frac{a+b}{2}$$

$$Var(X) = \int_a^b \frac{x^2}{b-a} dx - \left(\frac{a+b}{2}\right)^2 = \frac{(b-a)^2}{12}$$

Some Useful Distributions – Exponential

One parameter, "rate" λ:

$$f_X(x) = \lambda e^{-\lambda x}, \qquad x \geq 0$$

and 0 otherwise.

$$F_X(x) = 1 - e^{-\lambda x}, \qquad x \geq 0$$

and 0 otherwise.

$$E[X] = \frac{1}{\lambda}$$
$$Var(X) = 1/\lambda^2$$

We can see why the term "rate" is used for λ – think of X as measuring some length of time.

Example

Jobs arriving to a server have CPU time that is exponentially distributed with mean 140 msec. The CPU scheduling discipline is quantum-oriented so that a job not completing within 100 msec will be routed back to the tail of the queue of waiting jobs.

Find the probability that a job has to wait for a second quantum.

Memoryless Property

- Suppose that $X \sim Exp(\lambda)$
- Suppose that we know $X > b$. What is the probability that $X > a + b$? (This only makes sense if $a \geq 0$.)

$$P\{X > a + b | X > b\} = \frac{P\{\{X > a + b\} \cap \{X > b\}\}}{P\{X > b\}}$$

$$= \frac{P\{X > a + b\}}{P\{X > b\}}$$

$$= \frac{e^{-\lambda(a+b)}}{e^{-\lambda b}}$$

$$= e^{-\lambda a}$$

which is exactly the same as $P\{X > a\}$.

Memoryless Property

- This is known as the “memoryless” property, the future is independent of the past, given the present. Knowing that time b has elapsed gives no information about the future.
- Fact: the exponential distribution is the only continuous distribution with the memoryless property.
- Often used to model phenomena with this property: if we assume memoryless, then the underlying distribution is exponential.

Example

What is the probability that a job does not finish in the second quantum, given that it does not finish in the first quantum?